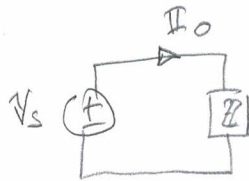


7.14)



BESTÄM P_f SÄMT LEAD/LAG

$$a) \begin{cases} V_s = 679 \cos(\omega t + 15^\circ) \text{ V} \\ I_o = 20 \cos(\omega t + 47^\circ) \text{ A} \end{cases}$$

$$P_f = \cos(\phi_v - \phi_i) = \cos(15^\circ - 47^\circ) = \cos(-32^\circ) \approx 0.848$$

$$\phi_v - \phi_i < 0 \quad \underline{\text{LEAD}}$$

$$b) \begin{cases} V_s = 163 \cos(\omega t + 15^\circ) \text{ V} \\ I_o = 20 \cos(\omega t - 22^\circ) \text{ A} \end{cases}$$

$$P_f = \cos(\phi_v - \phi_i) = \cos(15^\circ + 22^\circ) = \cos(37^\circ) \approx 0.798$$

$$\phi_v - \phi_i > 0 \quad \underline{\text{LAG}}$$

$$c) \begin{cases} V_s = 294 \cos(\omega t + 0^\circ) \\ I_s = 1.7 \cos(\omega t + 175^\circ) \end{cases}$$

$$P_f = \cos(\phi_v - \phi_i) = \cos(0^\circ - 175^\circ) = \cos(-85^\circ) \approx$$

$$\approx 0.0872 \quad \phi_v - \phi_i < 0 \quad \underline{\text{LEAD}}$$

$$d) Z_o = 48 + j16 \, \Omega$$

$$\phi = \text{ATAN}\left(\frac{16}{48}\right) = 18.43^\circ$$

$$P_f = \cos(\phi) = \cos(18.43^\circ) \approx \underline{0.949}$$

$$\phi_v - \phi_i > 0 \quad \underline{\text{LAG}}$$

SVAR: se ovan